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Day-Night Shifts in Water-Soluble Ions of Size-Resolved Aerosols before and after the COVID-19 Lockdown in a Coastal Megacity: Metro Manila, Philippines

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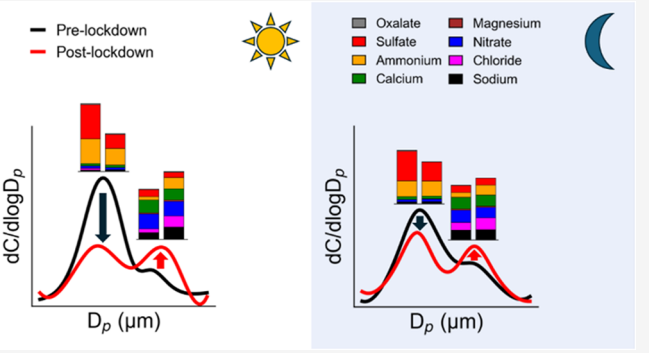
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ABSTRACT: The COVID-19 pandemic-driven lockdowns offer a unique opportunity to examine how reductions in anthropogenic emissions impacted atmospheric aerosol composition in urban environments. This study investigates the day-night variability of size-resolved water-soluble ions in ambient particulate matter (PM) collected in Metro Manila before (November 2019−February 2020) and after (November 2020−February 2021) lockdown implementation. Using tandem Micro-Orifice Uniform Deposit Impactors (MOUDIs), aerosol samples were collected during daytime (06:00−18:00) and nighttime (18:00−06:00) periods and analyzed for key ionic species (sulfate, ammonium, nitrate, oxalate, sodium, chloride, calcium, and magnesium) via ion chromatography. Submicrometer water-soluble mass declined post-lockdown, particularly during daytime, reflecting suppressed secondary formation under reduced anthropogenic activity, with substantial reductions in sulfate and ammonium. In contrast, concentrations in the supermicrometer range increased due to naturally higher sea salt levels. Chemical ratios reveal notable features post-lockdown and during daytime due to especially reduced sulfate levels: reduced chloride depletion (on percent basis), higher ammonium-to-sulfate ratios pointing to more excess ammonia available for reactions beyond neutralizing sulfate, and support for aqueous-phase processing preferentially forming more oxalate relative to sulfate. These findings underscore how both photochemistry and changes in anthropogenic activity influence aerosol composition, with implications for air quality and atmospheric processing in coastal urban cities.



KEYWORDS: *Metro Manila, size-segregated water-soluble ions, diurnal variation, chloride depletion, COVID-19*

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1. INTRODUCTION

Water-soluble aerosol species, which include both inorganic and organic ions, constitute a substantial fraction of particulate matter (PM) mass.[1,2] These species play a critical role in shaping aerosol properties and have wide-ranging impacts on air quality,[3] human health,[4] and climate through their influence on hygroscopicity, acidity, and radiative effects.[5]

The diurnal variability of water-soluble species depends on their chemical formation mechanisms as well as atmospheric conditions and sources that differ between daytime and nighttime. In urban environments, meteorological conditions and anthropogenic activities differ markedly over the course of the day.[6] Daytime is typically characterized by enhanced solar radiation that drives photochemical reactions that facilitate the oxidation of precursor gases (e.g., SO_2, NO_x, and VOCs), promoting the formation of secondary aerosol species.[6,7] In contrast, nighttime processes are influenced by shallower boundary layers, reduced photochemical activity, and alternative oxidation pathways such as reactions involving nitrate radicals.[8,9] Because these processes influence particles of different sizes in different ways, size-resolved measurements are essential to understand their atmospheric transformation, transport, and fate.[10,11]

The 2019 novel coronavirus disease (COVID-19) pandemic led to widespread reductions in anthropogenic emissions due to strict lockdown measures implemented globally. These changes, such as decreased vehicular traffic and reduced industrial activity, resulted in notable declines in air pollutants, including PM and NO_x, in many urban regions, e.g.,[12−14]

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including those within the Asian Monsoon regions.[15] However, the extent and duration of these reductions varied regionally due to differences in lockdown stringency and emission sources, e.g.,.[16,17] In Metro Manila, stringent restrictions beginning in March 2020 led to a decline in vehicular traffic and industrial operations, resulting in lower $PM_{2.5}$ levels compared to pre-lockdown conditions. As restrictions gradually eased, $PM_{2.5}$ concentration rebounded but still remained lower than pre-lockdown levels.[15] While most studies have focused on concentrations of broad classes of pollutants, fewer have examined how lockdown-related emission changes affected aerosol chemical composition, particularly size-resolved water-soluble species in coastal megacities such as Metro Manila.

Few studies have examined how potential compositional changes vary diurnally, highlighting the need to assess diurnal patterns in context of Metro Manila's complex air quality, which is primarily driven by anthropogenic sources such as vehicular traffic, industrial activities, and waste processing, along with other contributors including sea salt, aged aerosols, and biomass burning, e.g.,.[18−21] Prior studies at the same sampling site used in this work have characterized the size-resolved water-soluble ionic composition of PM using 48-h bulk samples. Aged aerosol and sea salt were identified as some of major sources of ionic PM in the region, with aged aerosol dominated by sulfate and ammonium.[20] Further work has shown that chloride depletion varies with pollutant type,[22] and that water-soluble species, particularly sulfate and ammonium, are more sensitive to rainfall events lasting more than an hour.[23] Variations in water-soluble species have also been associated with changes in aerosol composition linked to long-range transport and local emissions.[21]

Leveraging daytime and nighttime sampling across pre- and post-lockdown during COVID-19 pandemic, this study investigates the variability of size-resolved ionic species and gravimetric mass in Metro Manila, Philippines. The objective is to assess how water-soluble ions vary between both daytime and nighttime and also between pre-lockdown and post-lockdown periods. In doing so, special attention is given to a few ratios of interest related to sea salt reactivity and secondary formation mechanisms. Findings from this work provide insights into diurnal variability and pandemic-driven changes in aerosol composition, with broader implications for understanding atmospheric processing in coastal urban environments.

2. EXPERIMENTAL METHODS

2.1. Aerosol Sampling and Chemical Analysis. This study builds upon a previous measurement campaign conducted as part of the Cloud, Aerosol, and Monsoon Processes Philippines Experiment ($CAMP^2E_x$) weatHEr and CompoSition Monitoring (CHECSM) at the Manila Observatory (MO) site in Quezon City, Philippines (14.64° N, 121.08° E), which also employed Micro-Orifice Uniform Deposit Impactors (MOUDIs) to collect size-segregated aerosol samples over continuous 48-h periods.[24] The MOUDIs operated at ∼30 L/min with aerodynamic cut-point diameters (D_p) of 0.056, 0.10, 0.18, 0.32, 0.56, 1.0, 1.8, 3.2, 5.6, 10, and 18 μm.[25] In the present study, two MOUDIs worked in tandem with one sampling during daytime (06:00−18:00 local time) and the other at nighttime (18:00−06:00) to capture diurnal variability over three consecutive days, with total runtimes of 33−35 h per sample set. On selected days, the two MOUDIs were simultaneously used with one allocated for chemical analysis and the other for gravimetric analysis. For these sets, daytime filters were swapped out with nighttime filters when the nighttime period began and vice versa as soon as the daytime period started again the next day. The daytime period captures typical traffic rush hours, which occur from 07:00−09:00 and 16:00−18:00 local time.[26]

Sampling initially began in November 2019 and continued through February 2020. It was temporarily suspended due to the COVID-19 pandemic, then resumed in November 2020 as restrictions eased (Table S1), continuing until February 2021. The start and end dates for each MOUDI set are summarized in Table S2. In total, 36 sets (18 pairs of daytime and nighttime samples) were collected for ion chromatography (IC) analysis, and 12 sets (6 pairs) for gravimetric analysis.

The Teflon substrates were precleaned by soaking each side in Milli-Q water for at least 12 h, then dried and stored in methanol-cleaned Petrislides, and sealed with Parafilm to prevent contamination from particles or gases. Mass concentrations for each size range were determined by pre- and post-weighing the filters using a Sartorius ME5-F microbalance with a sensitivity of ±1 μg in a temperature and humidity controlled room. Each substrate designated for IC analysis was divided in half using ceramic scissors that were cleaned with methanol before and after each cut to avoid heavy metal contamination.[24] One half was extracted using 8 mL of Milli-Q water (18.2 MΩ·cm), followed by sonication and subsequent analysis for ionic species (Na^+, NH_4^+, Mg^{2+}, Ca^{2+}, oxalate, Cl^-, NO_3^-, SO_4^{2-}) using IC (Thermo Scientific Dionex ICS-2100 system). The other halves were preserved for future analyses. A more detailed discussion of the sampling procedures and analytical methods is provided in Stahl et al.[24]

2.2. Calculations and Data Analysis. The nonsea-salt fraction of sulfate ($nssSO_4^{2-}$) is determined using eqs 1−5, where SO_4^{2-} represents the total sulfate concentration. Both the ammonium-to-sulfate molar ratio and the oxalate-to-sulfate mass ratio, as well as the scatterplot shown in Figure 3 (Section 3.4), are based on $nssSO_4^{2-}$.The sea-salt sodium ($ssNa^+$) component is attributed to sea spray, and the constant 0.253 represents the SO_4^{2-}:Na^+ ratio characteristic of seawater.[27] Chloride depletion was calculated using eq 6, where 1.81 is the typical mass ratio of Cl:Na in seawater.[28] The mass concentration of lost Cl^- was quantified using eq 7.[29]

$$ssNa^+ = Na^+ - nssNa^+ \tag{1}$$

$$nssNa^+ = nssCa^{2+} \times \left(\frac{Ca^{2+}}{Na^+}\right)_{crust}; \text{ where } \left(\frac{Ca^{2+}}{Na^+}\right)_{crust} = 1.78 \tag{2}$$

$$nssCa^{2+} = Ca^{2+} - ssCa^{2+} \tag{3}$$

$$ssCa^{2+} = ssNa^+ \times \left(\frac{Ca^{2+}}{Na^+}\right)_{seawater}; \text{ where } \left(\frac{Ca^{2+}}{Na^+}\right)_{seawater} = 0.038 \tag{4}$$

$$nssSO_4^{2-} = SO_4^{2-} - (0.253 \times ssNa^+) \tag{5}$$

$$\%Cl^- \text{ depletion} = \frac{(1.81 \times [Na^+] - [Cl^-])}{(1.81 \times [Na^+])} \times 100\% \tag{6}$$

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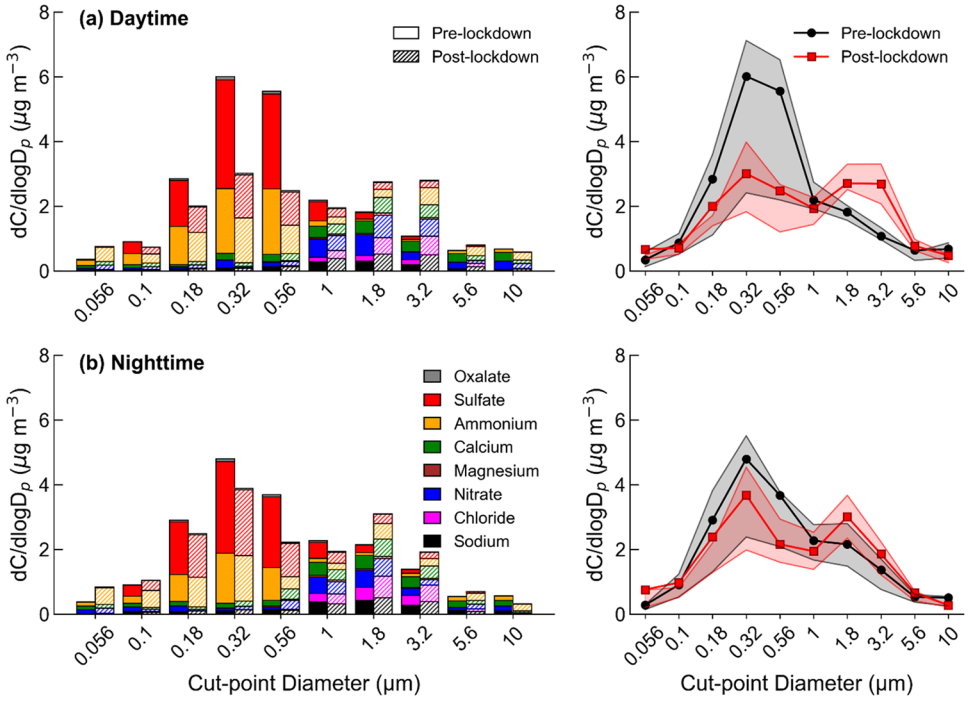


Figure 1. Mass size distributions of total water-soluble species (colored bars) for (a) daytime and (b) nighttime MOUDI sets. The hatched bars (i.e., always on the right of the pair of bars at same diameter bin) correspond to samples collected during COVID post-lockdown. The far-right panel in each subplot shows the mean mass size distribution profile of the total water-soluble species, with shading indicating the 25th to 75th percentile range.

$$\text{lost } Cl^- = ssNa^+ \times \left(\frac{Cl^-}{Na^+}\right)_{ss} - Cl^- \quad (7)$$

Daytime and nighttime samples were further categorized into two groups: pre-lockdown and post-lockdown. Pre-lockdown refers to samples collected before the onset of the COVID-19 pandemic (November 2019−February 2020), while post-lockdown refers to samples collected after lockdown restrictions were lifted but still during the pandemic period (November 2020−February 2021). During the pandemic, three different types of community quarantines were imposed depending on the COVID-19 risk levels.[30] Our post-lockdown period measurement fell under the General Community Quarantine (GCQ) classification (Table S1). Although public transport was already permitted during this period, most classes remained online, and remote work continued in some institutions (Table S1). As a result, anthropogenic emissions such as from vehicular transport, especially near the campus where sampling was conducted, were expected to be reduced due to continued remote arrangements. Additionally, because MOUDI set (MO5) was strategically conducted in a way to sample emissions from New Year's Eve celebrations, it was excluded from bulk analyses presented in Sections 3.2−3.4, but included in the overview discussion in Section 3.1.

3. RESULTS AND DISCUSSION

3.1. Cumulative Water-Soluble Ionic Composition. The total water-soluble mass concentration results are summarized in Table S2 ranging from 1.4−12.9 $\mu g\ m^{-3}$ during daytime and 2.2−10.2 $\mu g\ m^{-3}$ during nighttime. For the select sets when gravimetric analysis was conducted, the total bulk mass ranged from 7.4−25.6 $\mu g\ m^{-3}$ during the day and 19.1−38.9 $\mu g\ m^{-3}$ at night. During pre-lockdown, when comparing the contribution of water-soluble species to bulk mass across six gravimetric sets (3 daytime, 3 nighttime), the pre-lockdown period showed contributions ranging from 18.9−23.2% during daytime and 11.0−23.7% during nighttime. The elevated nighttime contribution of 23.7% was observed during a sampling period influenced by New Year's Eve celebrations (MO5-N), reflecting enhanced emissions of firework-related species that are ionic such as sulfate, nitrate, and chloride.[31] One of the lowest percentage contributions, 11.0% (MO4-N) was recorded during a period impacted by Typhoon Phanfone (category 2), which landed on 25 December 2019 and brought high amounts of rainfall (44.4 mm), and stronger winds compared to the other sets (Table S2). During the post-lockdown period, six gravimetric sets were also collected (3 daytime and 3 nighttime). The sums of water-soluble species accounted for 22.0−32.0% of the total gravimetric mass during daytime and 14.8−29.3% during nighttime, both of which were higher than their respective pre-lockdown values. For comparison, the water-soluble fraction was 33.4% of the total PM mass at the same sampling site between July−October 2018 using bulk 48-h sampling.[20]

3.2. Diurnal Variability. Figure 1 and Table S3 present the mean concentrations of water-soluble ions across different size fractions (0.056−1.0, 1.0−18.0, 0.056−18.0 μm) for both pre-lockdown and post-lockdown periods, separated into daytime and nighttime sampling. In the submicrometer range, sulfate was the most dominant species, followed by ammonium. During the pre-lockdown daytime, sulfate reached a mean concentration of 2.01 $\mu g\ m^{-3}$, which significantly decreased to 0.83 $\mu g\ m^{-3}$ post-lockdown. A similar trend was observed at night, with sulfate levels declining from 1.74 to 1.11 $\mu g\ m^{-3}$. These reductions are consistent with decreased SO_2 emissions from vehicles and industrial activities during the COVID-19 pandemic. Ammonium also showed notable reductions from pre- to post-lockdown during the daytime (1.42 to

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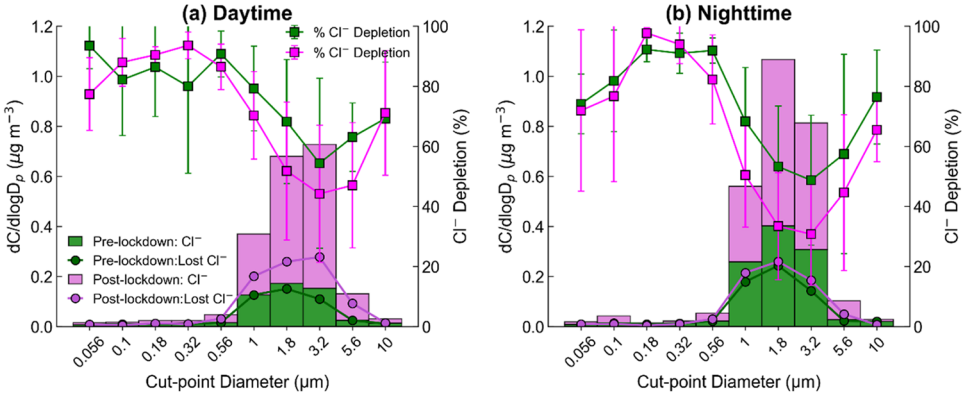


Figure 2. Average size distributions for Cl^-, Cl^- loss, and % Cl^- depletion during pre-lockdown and post-lockdown periods for (a) daytime and (b) nighttime samples. Error bars for % Cl^- depletion represent one standard deviation.

0.96 $\mu g\,m^{-3}$) with much less difference between nighttime levels (0.92 to 0.87 $\mu g\,m^{-3}$).

A higher NH_4^+:SO_4^{2-} molar ratio was observed post-lockdown (day/night = 6.12:3.71) compared to pre-lockdown levels (day/night = 3.41:2.21), indicating a relative increase in available ammonium compared to sulfate in the former period and especially during the daytime. This shift suggests more efficient neutralization of acidic aerosols and less acidic aerosol conditions, consistent with the reduced availability of strong acid precursors such as SO_2 and NO_x.[32] A previous study using 48-h MOUDI measurement at the same sampling site between July−October 2018 reported a mean NH_4^+:SO_4^{2-} ratio of 2.34 ± 0.52 for the submicrometer range.[20] Oxalate, which is a secondarily produced species from volatile organic carbon (VOC) precursor species (e.g., toluene, ethene, isoprene)[33,34] exhibited post-lockdown reductions in the submicrometer range as daytime concentrations decreased from 0.06 to 0.04 $\mu g\,m^{-3}$, while nighttime levels dropped from 0.05 to 0.03 $\mu g\,m^{-3}$. Oxalate is influenced by sources such as biomass burning,[35] combustion exhaust,[36,37] and biogenic sources.[36] A prior study at the same sampling site using 48-h bulk measurement further indicated contributions from crustal material and waste processing.[38]

In the supermicrometer range (1.0−18.0 μm), nitrate was the dominant species, followed by sulfate, calcium, sodium, and chloride. Nitrate showed no notable decrease in daytime levels post-lockdown, remaining nearly constant from 0.49 to 0.48 $\mu g\,m^{-3}$, whereas nighttime levels decreased only slightly more from 0.40 to 0.34 $\mu g\,m^{-3}$. Calcium decreased only slightly post-lockdown, with a reduction during daytime from 0.42 to 0.37 $\mu g\,m^{-3}$ and at night from 0.38 to 0.36 $\mu g\,m^{-3}$. Magnesium showed no difference between daytime and nighttime during pre- and post-lockdown (0.03−0.04 $\mu g\,m^{-3}$). In contrast, sodium and chloride both increased during the post-lockdown period. Daytime sodium concentrations increased from 0.22 to 0.40 $\mu g\,m^{-3}$, while nighttime levels increased slightly from 0.32 to 0.33 $\mu g\,m^{-3}$. Chloride concentrations increased more substantially, from 0.10 to 0.36 $\mu g\,m^{-3}$ during the day and from 0.25 to 0.39 $\mu g\,m^{-3}$ at night. A potential explanation for these increases are discussed further in the next section.

Daytime and nighttime differences in ionic concentrations were evaluated using the coefficient of divergence (CD)[39,40] and Wilcoxon signed-rank tests for each species and size fraction (Section S2). CD values above 0.2 indicated notable day-night heterogeneity during the pre-lockdown period for ammonium (both sub- and supermicrometer), chloride (both sub- and supermicrometer), and sodium (supermicrometer) (Table S4). Correspondingly, statistically significant day-night differences ($p < 0.05$) were observed pre-lockdown for sodium and chloride in the supermicrometer range and for chloride, magnesium, and ammonium in the submicrometer fraction. During post-lockdown period, only sulfate in the submicrometer range showed statistically significant day-night differences (Table S4).

Finally, the total water-soluble mass distributions shown in the right-hand panels of Figure 1 reveal a distinct post-lockdown decline in water-soluble aerosol mass, particularly within the submicrometer range. This reduction is especially pronounced during daytime periods, highlighting the sensitivity of secondary aerosol formation to local anthropogenic emissions and photochemical activity,[7,41] which is enhanced by high solar radiation, elevated temperatures, higher wind speed, and deeper boundary layer heights (Figure S1 and Section S1). In contrast, an enhancement in the supermicrometer range (1.8−5.6 μm), mainly attributed to increases in sodium and chloride concentrations, was observed post-lockdown, as indicated by a more distinct bimodal distribution during both daytime and nighttime.

3.3. Chloride's Anomalous Behavior. The previous section showed that, post-lockdown, only sodium and chloride increased in mass concentration, with chloride exhibiting the most substantial enhancement. As sea salt is the predominant documented source of these species in the study,[20,22] sea salt reactivity may explain the lower chloride concentrations pre-lockdown. This is because reduced anthropogenic emissions would result in depressed concentrations of acidic ions that would otherwise deplete more chloride in sea salt particles. Referred to as chloride depletion, this topic was previously examined at the same site using 48-h continuous MOUDI measurements by AzadiAghdam et al.[22] Building on that work, chloride depletion was quantified for samples collected specifically during daytime and nighttime, allowing for a more detailed investigation of temporal variations (Figure 2) as there are limited reports of how this process varies diurnally and especially when comparing pre- and post-lockdown periods in any region.

Chloride concentrations exhibited a clear peak in the coarse mode (1.0−5.6 μm) during both daytime and nighttime across the pre- and post-lockdown periods. Notably, nighttime Cl^- levels in the supermicrometer range were elevated post-lockdown compared to pre-lockdown (e.g., increasing from

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Table 1. Mean and Standard Deviation of Cl^- Depletion (%) and Cl^- Loss ($\mu g\ m^{-3}$) for All Size Range (0.056−18.0 μm), Submicrometer (0.056−1.0 μm), and Supermicrometer (1.0−18.0 μm)

diameter range (μm)	pre-lockdown (D)	pre-lockdown (N)	post-lockdown (D)	post-lockdown (N)
(a) Cl^- Depletion (%)				
0.056−18.0	75.8 ± 22.4	73.2 ± 23.8	70.7 ± 24.1	63.1 ± 29.9
0.056−1.0	85.8 ± 19.2	87.2 ± 11.6	87.5 ± 9.1	84.0 ± 21.6
1.0−18.0	66.7 ± 21.3	60.4 ± 24.8	56.6 ± 23.6	43.2 ± 22.1
(b) Cl^- Lost ($\mu g\ m^{-3}$)				
0.056−18.0	0.05 ± 0.09	0.07 ± 0.10	0.10 ± 0.15	0.09 ± 0.12
0.056−1.0	0.01 ± 0.01	0.01 ± 0.01	0.02 ± 0.02	0.02 ± 0.03
1.0−18.0	0.10 ± 0.12	0.13 ± 0.12	0.17 ± 0.17	0.16 ± 0.14

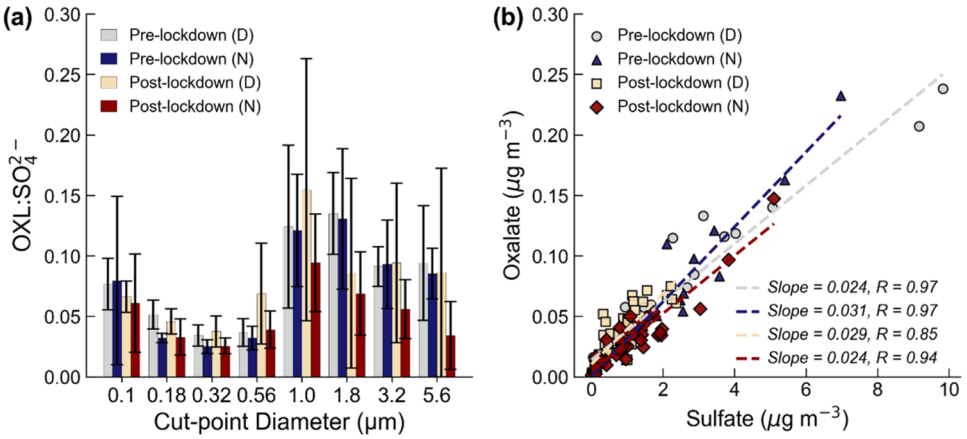


Figure 3. (a) Oxalate:SO_4^{2-} mass ratio as a function of particle cut-point diameter, and (b) linear regressions of oxalate and SO_4^{2-} mass concentrations in the submicrometer range (0.056−1.0 μm). Error bars in (a) represent one standard deviation. Only size bins with enough data points in the pre-lockdown (4−7 samples), and post-lockdown (7−10 samples) are shown in (a).

0.25 to 0.39 $\mu g\ m^{-3}$), suggesting either enhanced contributions from primary sea salt or reduced displacement of Cl^- by acidic species. Chloride depletion, which was most pronounced in the submicrometer range (0.056−1.0 μm), exceeded 80.0% across both periods, indicative of acid displacement reactions, such as those involving HNO_3 or H_2SO_4.[22,42] In contrast, depletion decreased with increasing particle size and was lower during the post-lockdown period, particularly at night, where values fell below 70.0% in the supermicrometer range (Table 1). This shift most likely reflects a reduction in atmospheric acidity post-lockdown due to decreased precursor emissions (e.g., NO_x and SO_2). Statistical analysis using the Wilcoxon signed-rank test revealed that the diurnal differences in Cl^- depletion were statistically significant only in the supermicrometer range during the post-lockdown period. However, when comparing pre- and post-lockdown, no statistically significant differences were observed.

While percent depletion is useful to report, it is just as important to quantify the absolute amount of Cl^- loss.[29] The submicrometer fraction showed relatively low values (<0.03 $\mu g\ m^{-3}$), with no significant day-night variability between pre- and post-lockdown periods. On the other hand, the supermicrometer range exhibited higher Cl^- mass loss, especially during the post-lockdown daytime (0.17 ± 0.17 $\mu g\ m^{-3}$), with values nearly double those observed in the daytime pre-lockdown period. This indicates that although depletion percentages were lower in the supermicrometer range, the absolute mass loss was more substantial. The reason that the absolute Cl^- loss is higher post-lockdown in the supermicrometer range even though the percent depletion was lower is that there were higher sodium (and thus sea salt) levels post-lockdown.

Overall, during the pre-lockdown period, Cl^- depletion was more pronounced during the day, while absolute mass loss was greater at night. In the post-lockdown period, both depletion percentages and mass loss were higher during daytime. While many past studies have reported reductions in various pollutants in urban areas due to changes in pandemic emissions rates, this study reports a change in sea salt reactivity and composition, with implications for sea salt's optical and hygroscopic properties.

3.4. Oxalate:Sulfate Ratio. Another chemical ratio of interest in this study is that between oxalate and sulfate (OXL:SO_4^{2-}), which is an indicator of aqueous-phase processing.[43,44] Hilario et al.[45] indicated that mass-based ratios between ∼0.015−0.03 are suggestive of air masses having undergone aqueous processing, while ratios greater than 0.3 are classified as high-OXL populations (e.g., coarse aerosol influence). In our data set, the OXL:SO_4^{2-} ratio did not exceed 0.3 on average, and the submicrometer range values were below 0.1 (Figure 3a). OXL:SO_4^{2-} ratios peaked in the supermicrometer range, indicating potential gas-particle partitioning of oxalate or its precursors onto coarse particles, consistent with previous observations at the same site using 48-h MOUDI sampling.[38,45] Additionally, linear correlations between oxalate and sulfate mass concentrations in the submicrometer range (Figure 3b) were strong across all periods (R = 0.85−0.97), indicating similarity in their formation pathways,[46] and most likely linked to aqueous processing.[34,47]

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Table 2. Mean and Standard Deviation of Oxalate:SO_4^{2-} Mass-Based Ratios for the Full Size Range (0.056−18.0 μm) and Submicrometer (0.056−1.0 μm) and Supermicrometer (1.0−18.0 μm) Ranges

diameter range (μm)	pre-lockdown (D)	pre-lockdown (N)	post-lockdown (D)	post-lockdown (N)
0.056−18.0	0.05 ± 0.01	0.04 ± 0.01	0.06 ± 0.03	0.04 ± 0.02
0.056−1.0	0.04 ± 0.01	0.03 ± 0.00	0.05 ± 0.02	0.04 ± 0.02
1.0−18.0	0.11 ± 0.04	0.12 ± 0.04	0.11 ± 0.07	0.07 ± 0.02

Table 2 shows numerical differences in this ratio between day and night and before and after lockdown periods. During both the pre- and post-lockdown periods, the submicrometer ratio is higher during the day compared to nighttime (pre: 0.04 vs 0.03; post: 0.05 vs 0.04), while there are higher overall ratios in the post-lockdown period due to systematically less sulfate as compared to oxalate (Table S3). This is because the predominant precursor to sulfate is anthropogenically derived SO_2, which was reduced post-lockdown, whereas oxalate is derived from VOCs that stem from more diverse sources including natural ones unaffected by the lockdown. The higher daytime ratios are suggestive of more favorable production of oxalate relative to sulfate, which warrants further investigation as more statistics from this and other regions would be beneficial.

For the supermicrometer range between 1.0−18.0 μm, only slight day-night difference was observed during pre-lockdown period (0.11 vs 0.12). In contrast, post-lockdown values show a higher daytime ratio compared to nighttime (0.11 vs 0.07), although nighttime value is lower than the pre-lockdown level. Thus, the partitioning behavior of these acids to coarse aerosol surfaces does not vary as much diurnally or between pre- and post-lockdown in contrast to their relative formation mechanisms in the submicrometer range.

4. CONCLUSIONS

This study provides new insights into the diurnal variability and pandemic-driven shifts in the chemical compositions of size-resolved aerosols in a Southeast Asian megacity. Water-soluble species contributed 18.9−23.2% to total gravimetric mass (<18 μm) during daytime and 11.0−23.7% during nighttime in the pre-lockdown period, increasing post-lockdown to 22.0−32.0 and 14.8−29.3%, respectively. A distinct decline in total water-soluble aerosol mass was observed post-lockdown for the submicrometer range, especially during the day, emphasizing the sensitivity of secondary aerosol formation to anthropogenic emissions and photochemistry. This decline was driven in part by reductions in sulfate and ammonium, likely due to decreased emissions of precursor gases from anthropogenic sources. The depressed levels of sulfate post-lockdown (especially daytime) resulted in shifts in aerosol composition in the form of reduced chloride depletion (on percent basis) and higher ammonium-to-sulfate ratios and thus more available ammonia for reactions such as neutralizing nitric acid. Also, while oxalate and sulfate were strongly correlated in the submicrometer range pointing to aqueous-phase formation, the oxalate-to-sulfate ratio was higher post-lockdown and especially daytime due to reduced sulfate precursors.

In contrast, there was an enhancement in water-soluble mass post-lockdown in the supermicrometer range driven by naturally higher sea salt levels (e.g., sodium and chloride). Although the percentage of chloride depletion decreased post-lockdown due to reduced acid levels, the absolute Cl^- mass loss increased, driven by higher sodium levels. Nitrate, calcium, and magnesium showed minimal changes, reflecting their stronger association with natural sources less affected by shifts in human activity.

Collectively, the results highlight the importance of considering both particle size and time of day when evaluating aerosol composition and atmospheric reactivity, particularly in regions with pronounced diurnal differences in human activity. These findings also demonstrate how large-scale societal disruptions, such as the COVID-19 pandemic, offer unique opportunities to investigate the sensitivity of aerosol composition to emission changes. Future studies with concurrent measurements of precursor gases and higher time resolution aerosol chemical composition are recommended to better understand the mechanisms driving aerosol changes in response to both emission reductions and meteorological variability.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsearthspacechem.5c00192.

Summary of meteorological parameters and a brief discussion of their daytime and nighttime profiles (Section S1 and Figure S1); coefficient of divergence (CD) calculation (Section S2); daytime and nighttime wind rose plots and HYSPLIT back trajectories (Figure S2); Metro Manila quarantine classifications (Table S1); summary of operating parameters and meteorological conditions for the MOUDI sample sets (Table S2); concentrations of water-soluble ions across different size ranges (Table S3); and summary of CD and Wilcoxon signed-rank test results (Table S4) (PDF)

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Data analysis was conducted by G.B. Writing of the original draft was performed by G.B. and A.S. All other authors contributed in part to the data collection, and the review and editing of the manuscript. All authors have approved the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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